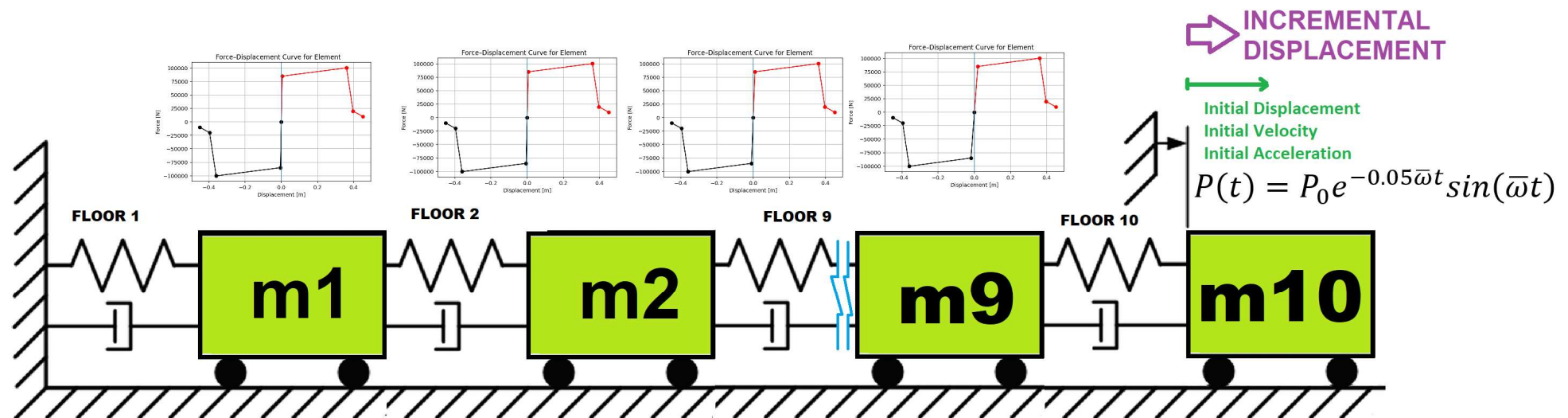


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

DETERMINATION OF OPTIMUM STRUCTURAL INITIAL ELASTIC STIFFNESS FROM STRUCTURAL PERIOD USING FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES WITH AND WITHOUT PARALLEL COMPUTING

THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

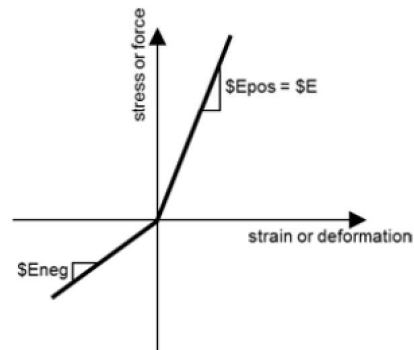
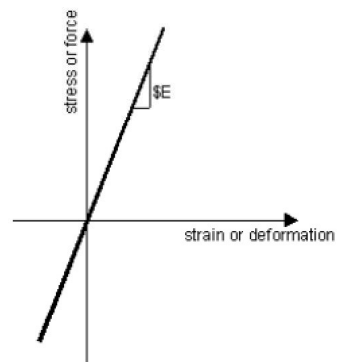
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

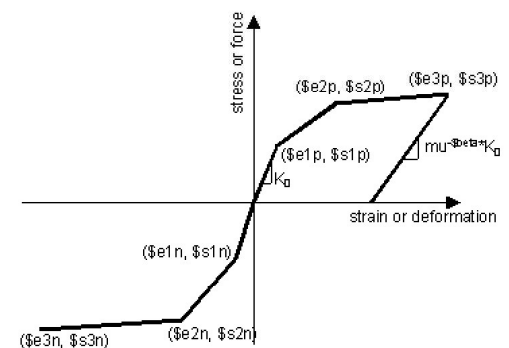
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

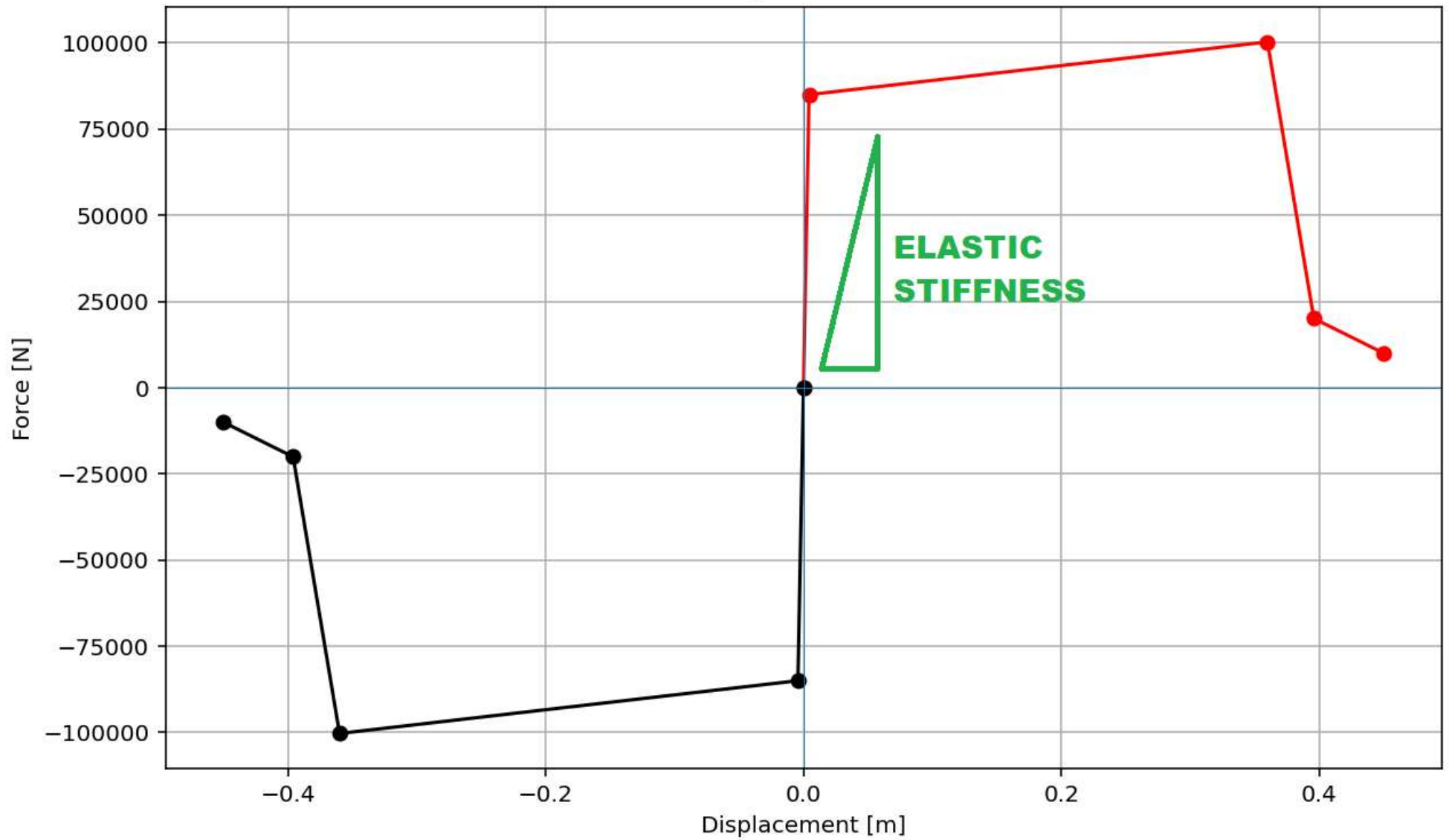


ELASTIC MATERIAL



INELASTIC MATERIAL

Force-Displacement Curve



**OPTIMIZATION
PROBLEM FOR
STRUCTURAL PERIOD
BY CHANGING INITIAL
ELASTIC STIFFNESS**

Spyder (Python 3.12)

FileEditSearchSourceRunDebugConsolesProjectsToolsViewHelp

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KE_OPTIMIZATION_PE...OORS_04_COLUMNS.py

PARA_KE_OPTIMIZATI...OORS_04_COLUMNS.py

1#####

2#>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

3#DETERMINATION OF OPTIMUM STRUCTURAL INITIAL ELASTIC STIFFNESS FROM STRUCTURAL PERIOD USING

4#FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES

5#-----

6#EQUIVALENT VISCOUS DAMPING RATIO: $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$

7#-----

8#ENERGY DISSIPATION CAPACITY INDEX = $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacement})$

9#-----

10#EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANALY

11#-----

12# $P(t) = P_0 \sin(\omega t)$

13# $P(t) = P_0 \exp(-0.05\omega t) \sin(\omega t)$

14#-----

15#ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION

16#OF ENERGY DISSIPATION CAPACITY INDEX

17#-----

18#MARKOV CHAIN FOR EVALUATIONS OF PROBABILITY OF FAILURE

19#-----

20#THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

21#EMAIL: salar.d.ghashghaei@gmail.com

22#####

23"""

24Nonlinear Seismic Performance Assessment of a MDOF:

25An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Earthquake Lc

26

27This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a MDOF

28for performance-based earthquake engineering. The 2D model incorporates both material nonlinearity

29(elastic-perfectly plastic or hysteretic steel with strain hardening)

30and geometric nonlinearity (corotational truss formulation) to capture P-delta effects

31and large displacements—essential for collapse assessment.

32

33Eight analysis protocols are implemented:

34(1) [PERIOD] : Structural Period

Console 1/A

1029 2029

1030 2030

1031 2031

1032 2032

1033 2033

1034 2034

1035 2035

1036 2036

1037 2037

1038 2038

1039 2039

+-----+-----+-----+-----+

| lambda | omega | period | frequency |

+-----+-----+-----+-----+

| 3.948e+03 | 62.8319 | 0.1000 | 10.0000 |

| 3.489e+04 | 186.7817 | 0.0336 | 29.7272 |

| 9.341e+04 | 305.6366 | 0.0206 | 48.6436 |

| 1.732e+05 | 416.1545 | 0.0151 | 66.2330 |

| 2.656e+05 | 515.3208 | 0.0122 | 82.0159 |

+-----+-----+-----+-----+

Fmax: -2.1246893133763933e-14

DF: -2.123301534595612e-11

DX: -0.0

IT: 31 - RESIDUAL: 0.0 X: 2334345758.395709

Optimum X (INITIAL ELASTIC STIFFNESS): 2334345758.3957

Iteration Counts: 31

Convergence Residual: 0.0000000000e+00

Total time (s): 0.5625

IPython Console

Plots

Variable Explorer

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Debugger

History

Files

Inline

Conda: anaconda3 (Python 3.12.7)

✓ LSP: Python

Line 1130, Col 14

UTF-8

CRLF

RW

Mem 38%

Fmax: -2.1246893133763933e-14

DF: -2.123301534595612e-11

DX: -0.0

IT: 31 - RESIDUAL: 0.0 X: 2334345758.395709

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Total time (s): 0.5625

**OPTIMIZATION PROBLEM
FOR STRUCTURAL
PERIOD BY CHANGING
INITIAL ELASTIC
STIFFNESS
(PARALLEL COMPUTING)**

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

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KE_OPTIMIZATION_PE...OORS_04_COLUMNS.py x PARA_KE_OPTIMIZATI...OORS_04_COLUMNS.py x

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # DETERMINATION OF OPTIMUM STRUCTURAL INITIAL ELASTIC STIFFNESS FROM STRUCTURAL PERIOD USING
4 # FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES
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6 # PARALLEL COMPUTING PROCESS
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15 #  $P(t) = P_0 \exp(-0.05\omega t) \sin(\omega t)$ 
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34
```

Console 1/A x

1.732e+05	416.1545	0.0151	66.2330
2.656e+05	515.3208	0.0122	82.0159

IT: 21 - RESIDUAL: 5.367326e-06 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 22 - RESIDUAL: 6.642312e-06 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 23 - RESIDUAL: 2.540489e-06 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 24 - RESIDUAL: 3.244646e-06 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 25 - RESIDUAL: 5.216824e-06 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 26 - RESIDUAL: 1.601025e-05 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 27 - RESIDUAL: 1.672026e-05 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 28 - RESIDUAL: 2.245749e-05 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 29 - RESIDUAL: 1.582070e-05 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 30 - RESIDUAL: 1.278364e-06 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
IT: 31 - RESIDUAL: 0.000000e+00 - INTIAL ELASTIC STIFFNESS: 2.334346e+09
Optimum Yield Strength: 2334345758.3957
Iteration Counts: 31
Convergence Residual: 0.0000000000e+00
Current time (HH:MM:SS): 11:27:00

In [5]:

IPython Console Plots Variable Explorer Help Debugger History Files

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1141, Col 23 UTF-8 CRLF RW Mem 38%

IT: 31 - RESIDUAL: 0.000000e+00 - INTIAL ELASTIC STIFFNESS: 2.334346e+09

Optimum Yield Strength: 2334345758.3957

Iteration Counts: 31

Convergence Residual: 0.0000000000e+00

Current time (HH:MM:SS): 12:03:41